Machine Learning and Computational Statistics
1st Homework

Ex 1

Let $x = [x_1\ x_2]^T$ be the input / feature vector and y the output variable in a regression task. Consider the following scenarios that relate y with x:

(i) $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + x_1^2$

(ii) $y = \text{sign}(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2)$

(iii) $y = 2x_1 + \text{sign}(3-7)x_2 + ReLU(3)\, x_1 x_2$

(iv) $y = \theta + \theta x_1 + \theta x_2 + \theta x_1 x_2$

(a) How many and which are the parameters involved in each scenario?

(b) In which of the above scenarios y has linear dependence with the involved parameters?

Note! $\rightarrow \text{sign}(z) = \begin{cases} 1, & z \geq 0 \\ 0, & \text{otherwise} \end{cases}$

$\rightarrow ReLU(z) = \begin{cases} z, & z \geq 0 \\ 0, & \text{otherwise} \end{cases}$

(a) (i) 4 parameters : $\theta_0, \theta_1, \theta_2, \theta_3$

(ii) 5 parameters : $\theta_0, \theta_1, \theta_2, \theta_3, \theta_4$

(iii) $\text{sign}(-4) = 0$, $ReLU(3) = 3$ $\rightarrow$ $y = 2x_1 + 3x_1 x_2$ — Zero parameters

(iv) 1 parameter : θ

(b) i, iv

Ex 2

Consider different ML tasks where the input x is 2-dim, i.e. $x = [x_1, x_2]^T$, and the output y is scalar. Assume also that in each case, a data set that consists of pairs $(y_i, x_i) \equiv (y_i, [x_{i1}, x_{i2}]^T)$, $i = 1, \ldots, N$, is available. Which of the following models are parametric and which are non-parametric?

1) $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2$

2) $y = \min(x_1, x_2)$

3) $y = ReLU(\theta_0 + \theta_1 x_1)$

4) $y = \sum_{i=1}^{N} \theta_i (x_{i1} x_1 - x_{i2} x_2)$

Note: $\min(x_1, x_2) = \begin{cases} x_1 & , x_1 \geq x_2 \\ x_2 & , \text{otherwise} \end{cases}$

1) Parametric

The number of learnable parameters $(\theta_0, \theta_1, \theta_2)$ is independent of the dataset of size N.

2) Non-Parametric

The model contains no parameters.

3) Parametric

The number of learnable parameters (θ_0, θ_1) remains fixed regardless of the dataset of size N

4) Non-Parametric

The number of parameters grows proportionally with the size N of the dataset.

Ex 3

(a) Define the parametric set of the quadratic functions $f_\theta : \mathbb{R} \to \mathbb{R}$ and give 2 instances of it. What is the dimensionality of θ?

(b) Define the parametric set of the 3rd degree polynomials $f_\theta : \mathbb{R}^2 \to \mathbb{R}$ and give 2 instances of it. What is the dimensionality of θ?

(c) Define the parametric set of the 3rd degree polynomials $f_\theta : \mathbb{R}^3 \to \mathbb{R}$ and give 2 instances of it. What is the dimensionality of θ?

(d) Consider the function $f_\theta(x) : \mathbb{R}^5 \to \mathbb{R}$, $f_\theta(x) = \frac{1}{1 + \exp(-\theta^T x)}$. Define the associated parametric set and give 2 instances of it. What is the dimensionality of θ?

(e) In which of the above cases f_θ is linear with respect to θ?

(a) $M = \{ f_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 \mid \theta \in \mathbb{R}^3 \}$

$\rightarrow f(x) = 1 + 2x + 3x^2$, $\theta = [1 \;\; 2 \;\; 3]^T$

$\rightarrow f(x) = x^2$, $\theta = [0 \;\; 0 \;\; 1]^T$

dim of θ : 3

(b) $M = \{ f_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2 + \theta_6 x_1^3 + \theta_7 x_2^3 + \theta_8 x_1^2 x_2 + \theta_9 x_1 x_2^2 \mid \theta \in \mathbb{R}^{10} \}$

→ $f(x) = x_1^3 + x_2$

→ $f(x) = x_1 x_2^2$

dim of θ : 10.

(c) $M = \{ f_\theta(x) = \sum_{i+j+k \le 3} \theta_{ijk} x_1^i x_2^j x_3^k \mid \theta \in \mathbb{R}^{20} \}$

→ $f(x) = x_1 x_2 x_3$

→ $f(x) = x_1^3 + x_2^3 + x_3^3$

dim of θ : 20

(Αριθμός όρων πολυωνύμου βαθμού d με n μεταβλητές : $\binom{n+d}{d} = \binom{3+3}{3} = 20$)

(d) $M = \{ f_\theta(x) = \frac{1}{1+\exp(-(\theta_1 x_1 + \dots + \theta_5 x_5))} \mid \theta \in \mathbb{R}^5 \}$

dim of θ : 5

→ $f(x) = \frac{1}{1+\exp(-x_1)}$, $\theta = [1 \ 0 \ 0 \ 0 \ 0]^T$

→ $f(x) = 0,5$, $\theta = [0 \ 0 \ 0 \ 0 \ 0]^T$

(e) a, b, c

Ex 4

Verify that for two l-dimensional column vectors $\theta = [\theta_1, \dots, \theta_l]^T$ and $x = [x_1, \dots, x_l]^T$ it holds : $(\theta^T x)x = (xx^T)\theta$

→ $\theta^T x = c = \sum_{i=1}^{l} \theta_i x_i$ → $c \cdot x_j = \left(\sum_{i=1}^{l} \theta_i x_i \right) x_j$

→ xx^T is an $l \times l$ matrix → $\sum_{i=1}^{l} (x_j x_i) \theta_i = \left(\sum_{i=1}^{l} \theta_i x_i \right) x_j$

position (j,i) : $x_j x_i$

=> $(\theta^T x)x = (xx^T)\theta$.

-0,3
0,1

Ex 5

Consider the vectors $x_n = [x_{n1}, \ldots, x_{nl}]^T$, $n = 1, \ldots, N$. Define the $N \times l$ matrix X and the N-dimensional column vector y as follows:

$$X = \begin{bmatrix} x_1^T \\ \vdots \\ x_N^T \end{bmatrix} = \begin{bmatrix} x_{11} & \cdots & x_{1l} \\ \vdots & & \\ x_{N1} & \cdots & x_{Nl} \end{bmatrix} \quad \text{and} \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

(a) Verify the following identities: $X^T X = \sum_{n=1}^{N} x_n x_n^T$ and $X^T y = \sum_{n=1}^{N} y_n x_n$

(b) What is the size of X, y, $X^T X$ and $X^T y$?

(c) Assume that a column vector of 1's is added in front of the column $\begin{bmatrix} x_{11} \\ \vdots \\ x_{N1} \end{bmatrix}$ of matrix X.

(i) What will be the changes in the dimensionality of the quantities in (b)?

(ii) Do the identities given in (a) still hold?

(a) $X = \begin{bmatrix} x_1^T \\ \vdots \\ x_N^T \end{bmatrix} = \begin{bmatrix} x_{11} & \cdots & x_{1l} \\ \vdots & & \vdots \\ x_{N1} & \cdots & x_{Nl} \end{bmatrix} \rightarrow X^T = [x_1 \ \cdots \ x_N]$

$$\Rightarrow \begin{cases} X^T X = [x_1 \ \cdots \ x_N] \begin{bmatrix} x_1^T \\ \vdots \\ x_N^T \end{bmatrix} = \sum_{n=1}^{N} x_n x_n^T \\ X^T y = [x_1 \ \cdots \ x_N] \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} = \sum_{n=1}^{N} y_n x_n \end{cases}$$

(b) X : $N \times l$ $\qquad$ $X^T X$: $l \times l$

y : $N \times 1$ $\qquad$ $X^T y$: $l \times 1$

(c) (i) X : $N \times (l+1)$ $\Rightarrow$ $\begin{cases} X^T X : (l+1) \times (l+1) \\ X^T y : (l+1) \times 1 \end{cases}$

(ii) Yes, matrix product rules don't get affected

Ex 6

A body moves on a straight line and performs a smoothly accelerating motion (we begin to study its motion at the time instance $t=0$). In the following table is given the velocity at certain time instances:

t (sec)	1	2	3	4	5
v (m/sec)	5.1	6.8	9.2	10.9	13.1

(a) Estimate the initial velocity and the acceleration of the body, based on the above measurements, utilizing the least squares error criterion.
(b) Write down the equation that expresses the velocity of the body as a function of time t.
(c) Estimate the velocity of the body at $t=2.3$

(a) $t=[1 \;\; 2 \;\; 3 \;\; 4 \;\; 5]^T$

$v=[5.1 \;\; 6.8 \;\; 9.2 \;\; 10.9 \;\; 13.1]^T$

$\theta=[v_0 \;\; a]^T$

$$X=\begin{bmatrix}1 & 1\\ 1 & 2\\ 1 & 3\\ 1 & 4\\ 1 & 5\end{bmatrix} \qquad y=\begin{bmatrix}5.1\\ 6.8\\ 9.2\\ 10.9\\ 13.1\end{bmatrix}$$

$$\underset{(2\times5)(5\times2)}{X^TX}=\underset{(2\times2)}{\begin{bmatrix}5 & 15\\ 15 & 55\end{bmatrix}} \qquad \underset{(2\times5)(5\times1)}{X^Ty}=\underset{(2\times1)}{\begin{bmatrix}45.1\\ 155.4\end{bmatrix}}$$

$$\det(X^TX)=5\cdot55-15^2=50\neq0 \Rightarrow (X^TX)^{-1}=\frac{1}{50}\begin{bmatrix}55 & -15\\ -15 & 5\end{bmatrix}=\begin{bmatrix}1.1 & -0.3\\ -0.3 & 0.1\end{bmatrix}$$

$$\hat{\theta}=(X^TX)^{-1}X^Ty=\begin{bmatrix}1.1 & -0.3\\ -0.3 & 0.1\end{bmatrix}\begin{bmatrix}45.1\\ 155.4\end{bmatrix}=\begin{bmatrix}2.99\\ 2.01\end{bmatrix}$$

$$\Rightarrow \begin{cases} v_0=2.99 \text{ m/s} \\ a=2.01 \text{ m/s}^2 \end{cases}$$

(b) $v(t) = 2.99 + 2.01t$

(c) for $t = 2.3$ sec, $v(2.3) = 2.99 + 4.623 = 7.613$ m/s